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Are You In Suspense?

Are you wondering what happened to your order or payment, even after a fortnight? But did you mention your name and address clearly while sending payment, and what the payment was for?

We get several payments every month without adequate instructions or even identity of the person sending the payment. As a result we are unable to take appropriate action on these, till we get a complaint - if at all.

At EFY, we want you to be our satisfied customer. So please do let us know in case your payment or instructions are not even acknowledged within a fortnight. We are not that lazy!

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EFYGROUP
Technology Drives Us

on top fell just short (0.78x), but the interconnects scaled further than expected (0.65x), thanks in part to the introduction of a new feature, airgaps, that reduces cross-talk between interconnects spaced so closely together. The result is that overall logic area scaling continues.

IP reuse and foundry business

All semiconductor companies develop intellectual property (IP) that they use in their own products or licence to others. For example, 3G and 4G technologies are not just for smartphones and tablets. Development of IP also helps semfab with development partners and with its growing foundry business.

Today, these customers are bringing their own designs using their IP, supplemented with technology, but eventually plan to offer their own IP to customers. As semiconductor manufacturing grows more difficult and expensive, and the number of players dwindles (from 18 a decade ago to just four today in advanced logic), manufacturing edge grows more compelling to other companies.

Enterprise IT hardware is still big and important, but other customers include Cloud service providers, telcos and technical computing. Growth drivers are Cloud architectures, software-defined networks and network-functions virtualisation for communications, high-performance computing and, of course, Big Data and analytics.

And it is getting an increasing share of its revenues from other data centre technologies including storage controllers and Ethernet switches, silicon photonics to replace fibre and copper, fabrics and switches and communications gear. The Internet of Things (IoT), software and services and memory sometimes get lost in the shuffle, but each is on track for more than US\$ 2 billion in sales this year.

Although gadgets like MICA

bracelet, Basis Peak smartwatch and SMS audio smart earbuds get lots of press, the IoT business is growing quickly due to applications such as retail, automotive and manufacturing.

The software business is growing more strategic and Intel has said, it believes that mobile devices based on its silicon will be the first—other than Nexus-branded ones—to get Android 5.0 Lollipop update.

The memory business, a joint venture with Micron, is shifting from selling memory chips to developing clients and enterprise SSDs, which are increasingly important to close the widening performance gap between CPUs and hard drive based storage.

Transistor options

You must select the right transistor architecture for 5nm. Today, there are two leading options: finFET and nanowire FET. Scaling finFET to 5nm is challenging. In a 5nm finFET, for example, fin width is projected to be 5nm, which is supposedly the theoretical limit for this structure. That is why chipmakers are exploring nanowire FET.

Nanowires have good electrostatics. But these come with other issues such as device width of nanowires and how much current you can get out of devices. Those are the areas that people are flushing out.

Mask making

In process flow, meanwhile, photomask manufacturing is one of the first steps. As before, lithography determines mask type and specs. So if 5nm happens, the photomask industry will likely need to develop masks for two lithography types—optical and EUV lithography. Making an optical mask will be daunting at 5nm. Bringing up EUV mask line is also difficult, as these are very different from 193 masks in many ways.